$$\min_{\mathbf{x}\in\mathcal{X}} \; c^{\top}\mathbf{x} \;+\; \max_{\mathbf{w}\in\mathcal{W}(\Gamma)} \; \min_{\mathbf{y}\in\mathcal{Y}(\mathbf{x},\mathbf{w})} \; d^{\top}\mathbf{y}$$

1 Invest
here-and-now: generation and storage capacity

2 Transmission blackout
adversary: worst outage window

3 Dispatch
wait-and-see: generation, storage, load shedding

Uncertainty set

$\mathcal{W}(\Gamma) = \{\mathbf{w} \in \{0,1\}^T : \; w_t = 1 \text{ on one contiguous block of } \Gamma \text{ hours}\}$

$w_t = 1 \;\Rightarrow$ **all** AC lines and DC links are out: every region becomes an island and must ride through on local generation and storage.

Capacity is committed before the timing is known; dispatch adapts once it is. No probabilities needed: the worst window is found endogenously.

Modelling the temporal transmission outage

